LECTURE - 17

Wednesday (online) 1/4/26.

Perceptron Learning Algorithm:

A mathematical process of training a "single layer" ANN to adjust its weights and bais since all the accuracy depends upon these.

Step 1: Short hand Rule (mistake only).

Initialize weights $= 0$, bias $= 0$.

Step 2 (IF y predicted $\neq$ y orignal then do)

CHANGE weight & biases:

learning rate

input

orignal output

$$\text{new } w = (\text{old } w) + (\eta \cdot x \cdot y)$$

$$\text{new } b = (\text{old } b) + \eta \cdot y)$$

Step 3: Repeat for all training data.

Step 4: Output Final weights & bias which gives all correct output.

⟹ Sign Activation Function.

predicted value

predicted y

$$f(y') = \begin{cases} 1 & y' > 0 \rightarrow \text{if } \Sigma b > 0;\ \text{sign} = 1 \\ -1 & y < 0 \\ 0 & y = 0 \end{cases}$$

Q: Find final weight and bais. $\eta = 0.5$, all bais/weight $= 0$

			linear combo.	Act. func	old			new		
x1	x2	y	y' / Σb	f(y')	w1	w2	b	w1	w2	b
① 0.5	0.2	-1	0	0 X	0	0	0	-0.25	-0.1	-0.5
② 1	0.5	-1	-0.8	-1	-0.25	-0.1	-0.5	same		
③ 6	0.8	1	-2.08	-1 X	-0.25	-0.1	-0.5	2.75	0.3	0
④ 8	0.9	1	22.27	1	2.75	0.3	0	same		

① $\Sigma b = y' = w_1 \cdot x_1 + w_2 \cdot x_2 + b$ // Linear Combo

$= 0 \times 0.5 + 0 \cdot 0.2 + 0$

$= 0$

$f(y') = \text{Sign}(0) = \boxed{0}$ // Activation func. sign gives predicted value $= 0$.

★ Since $f(y') = 0$ and y (orignal/expected) $= -1$;
∴ we need to change weights & find new ones for next row.

$$w1' = w1 + \eta(x1)(y)$$
$$w1' = 0 + (0.5)(0.5)(-1)$$
$$= \boxed{-0.25}$$
$$w2' = w2 + \eta(x2)(y)$$
$$w2' = 0 + (0.5)(0.2)(-1)$$
$$= \boxed{-0.1}$$
$$b' = 0 + 0.5(-1) = \boxed{-0.5}$$

② $y' = \Sigma b = (-0.25)(1) + (-0.1)(0.5) + (-0.5)$
$$= \boxed{-0.8}$$
$f(y') = \text{Sign}(-0.8) = \boxed{-1}$

★ Since $f(y') = y = -1$ ∴ No update needed

③ $y' = (-0.25)(6) + (-0.1)(0.8) + (-0.5)$
$$= -2.08$$
$f(y') = \text{Sign}(-2.08) = \boxed{-1}$

★ Update in weights & bias needed as $y \neq f(y')$

$$w1' = (-0.25) + (0.5)(6)(1)$$
$$= \boxed{2.75}$$
$$w2' = (-0.1) + (0.5)(0.8)(1)$$
$$= \boxed{0.3}$$
$$b' = -0.5 + (0.5)(1) = \boxed{0}$$

④ $y' = 22.27$, $f(y') = +1$, ∴ Matched with 'y' (no updated).

NOTE: Now since row ① & ③ does not had $f(y')$ (predicted) same as y (expected output) ∴ we'll do another iteration with the entire four row dataset with $w1 = 2.75$, $w2 = 0.3$, $b = 0$ UNTIL all four rows has $f(y') = y$.
These wld be final $\boxed{w1, w2, b}$.

I Standard Rule (error - Based):

* Same steps as the "Short Hand Rule"; there is just a different formula used for new bias & weights ~~for each row.~~

~~But even if $f(y') = y$ which means predicted = actual; we'll still update weight & bias~~

$$\text{new } w = (\text{old } w) + \eta \underbrace{(y - f(y'))}_{\text{error} = \text{actual output} - \text{predicted}} \cdot x$$

$$\text{new } b = (\text{old } b) + \eta \, (y - f(y')) \cdot x$$

* even if one row has different y and $f(y)$; we'll do another iteration just like prev. method.